East West University

Department of Mathematics and Data Science

Mid-term Examination (Summer 2026)

Course Title: **Linear Algebra and Complex Variables**

Course Code: **MAT 205** **Section: 8**

Course Instructor: Md. Ashraful Alam (Alam)

Total Marks: 30 *Time: **90 mi***

Answer all the questions.

1(a). Write down the conditions for the addition and multiplication of two matrices.

If $A = \begin{pmatrix} 1 & 0 & -2 \\ 2 & 2 & 4 \\ 0 & 0 & 2 \end{pmatrix}$ and $f(x) = x^2 - 3x + 2$, then find $f(A)$.

(b) Determine the cofactor of each entry of the following matrix

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 1 & 0 & 2 \\ 4 & 1 & 5 \end{pmatrix}.$$

(c) Distinguish between Row Echelon Form (REF) and Reduced Row Echelon Form (RREF). Can a matrix have more than one Reduced Row Echelon Form? Justify your answer. 5

Find the Rank of the matrix $A = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 2 & 1 & 3 & 5 \\ 3 & 1 & 4 & 7 \\ 4 & 2 & 6 & 10 \\ 5 & 2 & 7 & 12 \end{pmatrix}$.

2(a). Briefly discuss the advantages of the **Gauss Elimination Method** over Cramer's Rule. Solve the following system of linear equations using **Cramer's Rule or Gauss Elimination Method**: 4

$$x - 4y + z = 6$$
$$4x - y + 2z = -1$$
$$2x + 2y - 3z = -20$$

(b) i) Explain why every homogeneous system of linear equations is always consistent.
ii) A homogeneous system has four equations and four unknowns with a non-zero determinant of the coefficient matrix. Predict the nature of its solution. Explain your reasoning.

iii) Find the general solution of the following homogeneous system of linear equations using Gauss-Jordan elimination method. Also, determine two particular solutions of this system.

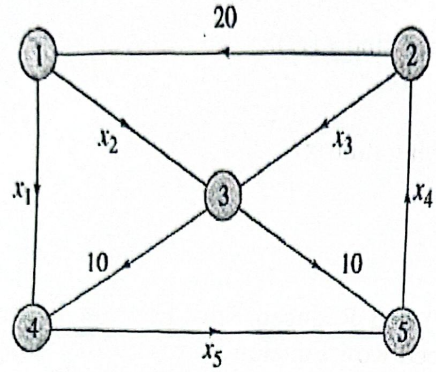
$$x + y + z + w = 0$$
$$2x + 2y + 2z + 2w = 0$$
$$x - y + z - w = 0$$
$$3x + y + 3z + w = 0$$

) Use the LU-decomposition $\begin{pmatrix} 3 & -6 & -3 \\ 2 & 0 & 6 \\ -4 & 7 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 2 & 4 & 0 \\ -4 & -1 & 2 \end{pmatrix}\begin{pmatrix} 1 & -2 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$

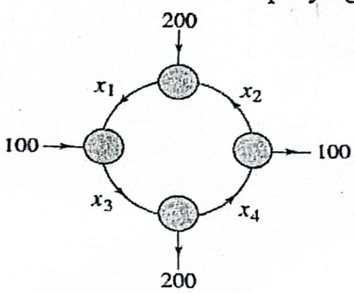
solve the system

$$3x - 6y - 3z = -3$$
$$2x + 2z = -22$$
$$-4x + 7y + 4z = 3$$

3(a) Set up a system of linear equations to represent the network shown in the fol figure.



(b) Form a system of linear equations for the flow of traffic (in vehicles per hour) through a network of streets is shown in the accompanying figure.



Find the traffic flow when $x_4 = 100$.

(c) The expression levels of three genes (G_1, G_2, G_3) under three environmental conditions (C_1, C_2, C_3) are represented by:

	G_1	G_2	G_3
C_1	12	8	5
C_2	6	15	7
C_3	9	4	11

Determine:

a) Total expression of each gene
b) Total expression under each condition
c) Average expression of each gene
d) The most highly expressed gene

ID No.: ....